ODWA NOMBAMBELA

Computer Engineer
IoT & Embedded Systems

CONTACT

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SKILLS

- C / C++, STM32, Embedded Firmwa
- MQTT, LoRa, AT Commands, CAT-1 LTE
- React / Vite, Node.js, Express
- PostgreSQL, AWS RDS, Docker
- Python (NumPy, OpenCV, pyotdr)
- OTDR Analysis & Fibre Splicing
- Solar Power System Design
- Git, Linux, LaTeX

LANGUAGES

English	**Fluent**
IsiXhosa	**Fluent**
IsiZulu	**Intermediate**
Afrikaans	**Basic**

INTERESTS

- Programming & Open Source
- Fibre & RF Infrastructure
- Photography
- Squash & Rugby
- Technical Reading

PROFILE

Newly graduated Computer Engineer (BEng, University of Pretoria - qualification obtained November 2025, graduated May 2026) with a strong foundation across the full engineering stack: from low-level C firmware on STM32 microcontrollers to cloud-connected IoT platforms and React/Node.js development. Currently working at Take Note IT (TNIT) as an R&D engineer, contributing to live infrastructure protection deployments in the City of Ekurhuleni. Proven ability to independently design, debug, and deliver complex systems under real-world field constraints. Equally comfortable writing interrupt-driven firmware, analysing OTDR traces on live fibre links, or architecting PostgreSQL-backed web platforms on AWS.

WORK EXPERIENCE

R&D Engineer

Take Note IoT (TNIoT) — *January 2025 – Present*

- Developed base station firmware in C on the STM32F030C8T6 MCU, integrating RFM95W LoRa (868 MHz), LIS2DW12 accelerometer, and Air780EU CAT-1 LTE module.
- Resolved full firmware debugging cycle: UART clock errors, AT command parsing, LTE registration, MQTT transaction ordering, ISR race conditions, and RTC alarm re-arming edge cases.
- Optimised node deep-sleep current toward 15 µA by disabling SPI1/ADC1 clocks, fixing floating UART RX pins, and driving GSM_ENABLE pin LOW.
- Built the *Aware* IoT platform (React/Vite, Node.js/Express, PostgreSQL on AWS RDS, AWS Cognito, MQTT) replacing the legacy alerting stack; Dockerised for AWS deployment.
- Conducted OTDR testing and fibre fusion splicing on live single-mode links; built a Python PDF-report tool using pyotdr & ReportLab.
- Designed solar-powered remote base stations (100 W panel, SLA/LiFePO4 batteries) with battery-autonomy calculations for South African conditions.
- Deployed and maintained a Telegram monitoring bot on a Jetson Nano tracking ~15 base stations across the East Rand.

IT Support & Administrative Assistant

NS. Nombambela Inc — *Jan 2022 – Dec 2022*

- Managed technology infrastructure and IT consultation; handled front desk operations and document management systems.
- Handled front desk operations including customer inquiries, appointment scheduling, and administrative coordination.
- Maintained document management system and ensured efficient office operations and communication workflows.
- Provided technical troubleshooting and support for office hardware and software systems.

Tutor - Professional Orientation (JPO110 & JPO120)

University of Pretoria — *Feb 2017 – Nov 2017*

- Assisted lecturers during lectures, and mentored students through one-on-one consultations.
- Evaluated and marked student assignments, providing detailed feedback on performance.

Community Based Project Volunteer (EBIT WEEK - JCP203)

University of Pretoria — *June 2018 - July 2018*

- Provided guidance to prospective students and answered questions about the university programs and campus life.
- Conducted campus tours for prospective students and parents while facilitating communication between current students, university representatives and soon to be students.

KEY TECHNICAL PROJECT

Automated 3D Plant Phenotyping System

Final Year Project (EPR402), Univ. of Pretoria *Feb – Nov 2025*

- Designed a complete automated system for non-destructive plant biomass estimation using dual Kinect V2 RGB-D cameras on a custom 6×3×2 m gantry; achieved **93.4%** accuracy.
- 5-stage pipeline: preprocessing, sequential ICP registration (RMSE 14.1 mm), 3D reconstruction, dimensional analysis, and a 6-feature Random Forest model.
- Integrated ODROID N2+ with Arduino Uno motor controllers, NEMA 23 steppers, and TB6600 drivers via TCP sockets; delivered results within 2 minutes via a multi-threaded tkinter GUI.

EDUCATION

BEng Computer Engineering *May 2026*

University of Pretoria

Qualification obtained: 28 November 2025 | Certificate received: 14 May 2026

Relevant Coursework: Embedded Systems, Computer Vision, Machine Learning, DSP, Microcontrollers, Software Engineering, Data Structures & Algorithms

Short Course: I-SET Robotics Components and Pedagogy *2022*

University of South Africa (UNISA)

Matric Certificate *2015*

Selborne College

CERTIFICATIONS & TRAINING

- CISCO CCNA
- I-SET Robotics Components and Pedagogy - UNISA (2022)
- First Aid Level 3
- Ajax Systems Baseline Intrusion Course (2026)

REFERENCES

Mr Andile Baleni - Associate, NS. Nombambela Inc | Cell: 082 432 6403

Mr Tony Yard - Landlord | Cell: 082 374 4421

Mr Devique Barkley - CEO of Take Note IoT | Cell: 076 730 6720